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Chapter · October 2016

DOI: 10.1002/9781118766804.wbiect144

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Copenhagen Business School

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Kjeld Schmidt
Copenhagen Business School
schmidt@cscw.dk

Abstract: Computer-Supported Cooperative Work (CSCW) is an interdisciplinary research field devoted to understanding cooperative work practices with the aim of contributing, both conceptually and technically, to the development of collaborative computing, i.e., computing technologies that facilitate, mediate, or regulate workers' interdependent activities. The disciplinary span of CSCW as an interdisciplinary field is exceptional in that it bridges from ethnographic field work, to the experimental development of advanced technical solutions, to evaluation studies *in situ*.

Computer-Supported Cooperative Work (CSCW) is an interdisciplinary research field devoted to understanding cooperative work practices with the aim of contributing, both conceptually and technically, to the development of 'collaborative computing', i.e., computing technologies that facilitate, mediate, or regulate workers' interdependent activities.

The disciplinary span of CSCW as an interdisciplinary field is exceptional in that it bridges from ethnographic field work, to the experimental development of advanced technical solutions, to evaluation studies *in situ*, and does so in a systematic manner. (For an overview of CSCW research to date, cf. Bannon and Schmidt 2013).

Collaborative computing was not invented by CSCW but is rather a precondition for its formation. In fact, collaborative computing is as old as computing technology in general. The very first electronic computers were constructed during and immediately after World War II in order to automate the massive calculation work required in complex areas such as cryptology or the development of new weapon systems, from artillery to nuclear devices, which until then had been performed by 'human computers', and electronic computers were initially designed to eliminate the increasingly crippling overhead effort of planning and coordinating this work. However, already in the late 1940s another form of computing emerged, which we today would categorize as *collaborative computing*. It was not designed to automate calculation work and thereby eliminate coordination costs but rather to *assist* workers in their work and in the coordination of their work. Initially developed for US air defense (the SAGE system), collaborative computing developed into the technology generally known by the technical term 'online transaction processing'. Well-known examples are multiuser database systems such as airline reservation systems, online banking systems, etc. It is collaborative computing by virtue of facilitating cooperative

work by providing a shared digital representation, but it only offers workers very restricted means of coordinating their activities: when faced with contingencies, workers have to interact outside of the system, for instance, by means of conversations.

Later, with the invention of ‘time-sharing’, which gave multiple workers concurrent access to the same ‘host’ computer, technicians building these systems took the next step and built programs that allowed them to post messages to one another. However, they could only do so within the closed network of terminals connected to a host computer; but with the emergence of open computer networks in the 1970s and 80s (the Internet), communication technologies based on message-handling, such as electronic mail, instant messaging, chat rooms, etc., developed rapidly. So, while the early form of collaborative computing only provided rudimentary means of coordinating cooperative work, the new form of collaborative computing, often termed ‘computer-mediated communication’, provided means for ‘informal’ exchange of free-form messages, documents, etc.

However, computer-mediated communication technologies were soon found fundamentally inadequate for ordinary cooperative work. In the words of the CSCW pioneer Irene Greif (1988):

‘The unstructured body of messages is suitable for the free-flowing text of natural language, but does not let us set the computer to work on our problems. Designers who draw pictures, software developers who jointly write code, financial analysts who collaborate on a budget — they all need coordination capabilities as an integral part of their work tools. That means coordination support within the CAD engineer’s graphics package, within the programmer’s source-code editor, within the budget writer’s spreadsheet program. It means support for managing versions of objects, be they pictures, programs, or spreadsheets. It means ways to distribute parts of the object for work by contributing group members, ways to track the status of those distributed parts, ways to pull completed objects back together again. The limit of electronic mail and computer conferencing is that they have such features for managing messages only. CSCW widens the technology’s scope of application to all the objects we deal with.’ (Greif 1988, pp. 7 f.).

CSCW emerged in the 1980s as a research area from the insight that existing computational technologies fell short in the face of the problems of coordinating interdependent work activities involving multiple actors, especially when working under conditions of contingency and uncertainty. What was realized, by Greif and many others, was that collaborative computing raised a set of fundamental issues that are all connected to the nature of *cooperative work practices* and that, in order to be able to contribute to technologies that can be integrated into actual cooperative work practices, it was necessary for researchers to build their designs on a rich and sophisticated understanding of such practices (e.g., Suchman 1987; Schmidt and Bannon 1992). The feasibility of such an approach was demonstrated quite early in the history of CSCW by a few studies that have become paradigmatic exemplars (e.g., Harper et al. 1991; Heath and Luff 1992).

Central to the CSCW research program that resulted from these reflections is an approach to technology development radically different from that of the previous development of collaborative computing. It is an approach to technology

development in which ethnographic and other forms of in-depth workplace studies play an essential and proactive role (for discussions of the role of ethnography in CSCW, cf., e.g., Luff et al. 2000). It is thus a program that differs in fundamental ways, for example, from studies of computer-mediated communication (from email to social media) that typically are reactive in that they focus on anticipating possible organizational and social advantages or disadvantages of some already existing technology.

In line with this program, CSCW researchers, endeavoring to understand cooperative work and coordinative practices by investigating the manifold world of cooperative work, have over the years been building a large and rich corpus of workplace studies in a wide range of settings. In this effort, cooperative work in health care and the problems with electronic patient records have been subjected to intense study. Another domain that has attracted significant attention in CSCW is what can loosely be termed 'design work'. Within this domain, software engineering as a cooperative endeavor has, not surprisingly, also been studied systematically, and so have related domains of professional design work such as architectural work and construction. On the other hand, a distinctly different body of domain-oriented studies have investigated 'centers of coordination' such as rapid urban transit control rooms, air-traffic control, airport ground operations, ambulance dispatch centers, call centers, etc. Less intensely but nevertheless with significant findings, a vast spectrum of domains of cooperative work have been investigated, such as, for example, studies of process control work: space mission planning and control, waste water treatment, oil and gas production, as well as vehicle control, or at the other end of the spectrum, a large number of studies of cooperative work in financial institutions such as securities trading and retail banking.

In this effort of ethnographic and similar studies, CSCW has developed important elements of the conceptual framework that is required for it to contribute constructively to the development of collaborative technologies. Thus, complementing these studies, each of which focuses on specific work domains, a number of workplace studies are undertaken to investigate more general aspects of cooperative work such as, for example, 'diagnostic work', 'invisible work', 'mobility work', or 'nomadic work'. Orthogonal to studies focusing on work domains or types of work, but building on the evolving corpus of workplace studies, CSCW researchers have, very tentatively, developed conceptualizations of key aspects of cooperative work practices such as the temporal and spatial aspects of coordinative practices.

More specific to cooperative work practices, CSCW researchers are exploring categories of *coordinative practices*.

On one hand, a strand of study emerged quite early as ethnographic studies demonstrated that the performance of competent workers in their cooperative activities is characterized by 'effortless' coordination, i.e., the phenomenon that practitioners in conducting their own work routinely heed the ongoing activities of

their colleagues. The implications of these findings were quickly understood as having radical implications for collaborative technologies (Rodden et al. 1992).

On the other hand, as the program of ethnographic studies unfolded, it became evident that workers in complex cooperative work settings typically develop coordinative practices (plans, procedures, schedules, classification systems) that often are highly sophisticated and in which specialized coordinative artifacts play an essential role. A critical research effort is therefore devoted to trying to describe and understand the semiology of these artifacts in the light of the logic of the specific coordinative practices.

CSCW was formed as researchers — in academia, in governmental and business research labs, and in practical software engineering — began to realize that the problems they were trying to come to grips with might be similar to the problems other researchers were facing. This process of convergence is ongoing, as researchers faced with related problems arising from the design, implementation, and use of new technologies or types of system realize that their problems are already addressed in some relevant way or could benefit from being approached from a CSCW perspective. That is, driven by the introduction of new technologies, by innovative uses of known technologies, or by the application of collaborative technologies in new domains, CSCW is continually being formed as an interdisciplinary research field. This is a source of fragmentation but certainly also a source of intellectual energy, as manifested in the cumulative development, in an ongoing dialogue between computer scientists and social scientists from sociology, anthropology, social psychology, of the conceptual framework required to base the development of collaborative computing as a systematically elaborated and empirically grounded technology (for examples of contributions of this kind, cf., e.g., Heath and Luff 2000; Schmidt 2011).

SEE ALSO: Anthropology; Computer-mediated communication; Programming (computers); Psychology, social; Sociology; Technology; Theory and practice

References and further reading

- Bannon, L. J.; and Schmidt, K. (eds.) (2013). CSCW: The first quarter century. Special issue, *Computer Supported Cooperative Work (CSCW)*. 22 (4-6), pp. 345-715.
- Greif, I. (1988). Overview. In I. Greif (ed.): *Computer-Supported Cooperative Work: A Book of Readings*. San Mateo, Calif.: Morgan Kaufmann Publishers, pp. 5-12.
- Harper, R. H. R.; Hughes, J. A.; and Shapiro, D. Z. (1991). Harmonious working and CSCW: Computer technology and air traffic control. In J. M. Bowers and S. D. Benford (eds.): *Studies in Computer Supported Cooperative Work: Theory, Practice and Design*. Amsterdam: North-Holland, pp. 225-234.
- Heath, C. C.; and Luff, P. (1992). Collaboration and control: Crisis management and multimedia technology in London Underground control rooms. *Computer Supported Cooperative Work (CSCW)*. 1 (1-2), pp. 69-94.
- Heath, C. C.; and Luff, P. (2000). *Technology in Action*. Cambridge: Cambridge University Press.

- Luff, P.; Hindmarsh, J.; and Heath, C. C. (eds.) (2000). *Workplace Studies: Recovering Work Practices and Informing System Design*. Cambridge: Cambridge University Press.
- Rodden, T. A.; Mariani, J. A.; and Blair, G. (1992). Supporting cooperative applications. *Computer Supported Cooperative Work (CSCW)*. 1 (1-2), pp. 41-68.
- Schmidt, K.; and Bannon, L. J. (1992). Taking CSCW seriously: Supporting articulation work. *Computer Supported Cooperative Work (CSCW)*. 1 (1-2), pp. 7-40.
- Schmidt, K. (2011). *Cooperative Work and Coordinative Practices: Contributions to the Conceptual Foundations of Computer-Supported Cooperative Work (CSCW)*. London: Springer.
- Suchman, L. A. (1987). *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge: Cambridge University Press.

Biography

Kjeld Schmidt is Professor of Work, Organization, and Technology at Copenhagen Business School. He has been actively involved in the organization of the CSCW community for 25 years and has been the Editor-in-Chief of *Computer Supported Cooperative Work: The Journal of Collaborative Computing and Work Practices* since 1992. His main scholarly contributions to the field of CSCW are centered on what can be termed its conceptual foundations. His research is particularly aimed at uncovering the logics of coordinative practices and the associated artifacts.